Examiner only

2. Brewing is an important industry in the UK.

(a) Complete the word equation for the fermentation of beer by underlining the correct word(s) in the brackets. [1]

glucose ⟶ ethanol + **(carbon dioxide / oxygen / nitrogen)**

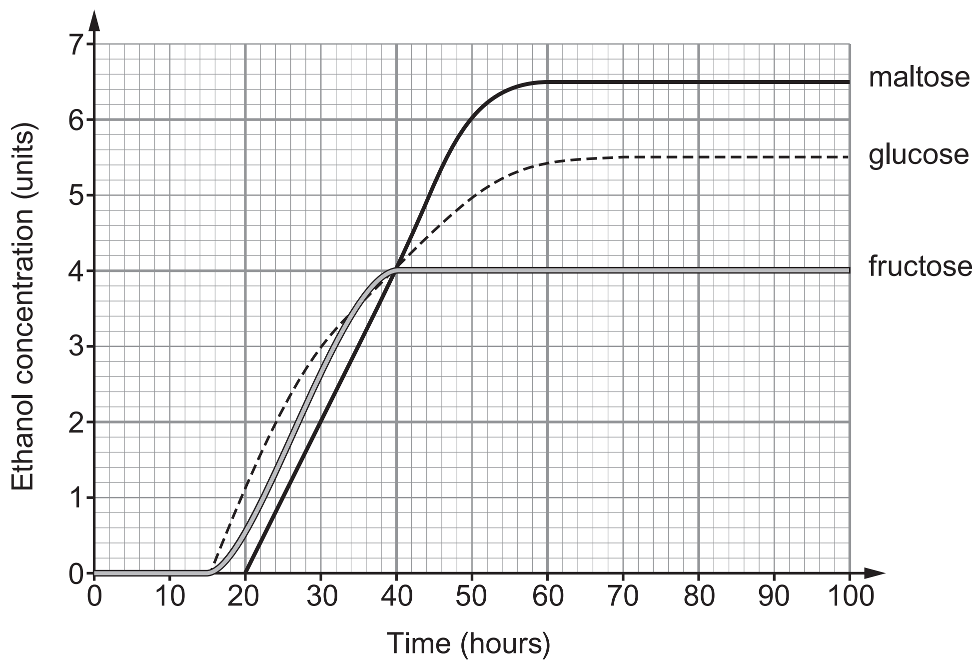
(b) The main stages of the beer making process are shown below. However, the stages are not in the correct order. [2]

A	starch in the barley is converted to sugar, and water is added to make a brewing mixture
B	fermentation stops when all the sugar runs out
C	mixture is cooled and yeast is added
D	mixture is boiled
E	the beer is pasteurised, then stored in bottles or cans
F	hops are added for flavour

Put the statements in the correct order.

A	**F**				**E**

(c) A brewery tested different types of sugar in the fermentation process. These were glucose, maltose and fructose.
The results are shown in the graph below.



Examiner only

Use information from the graph and the numbers in the box to answer the following questions.

2	**3**	**6**	**40**	**50**	**70**

(i) State the concentration of ethanol formed from **glucose** at **30 hours**. [1]

.... units

(ii) State the time at which the concentration of the **ethanol** was the same for all the sugars. [1]

time = ... hours

(iii) State the time at which fermentation stops for **glucose**. [1]

time = ... hours

(iv) Calculate the difference in ethanol concentration between 20 and 50 hours for maltose. [1]

concentration difference = ... units

(d) (i) Use your answer in (c)(iv) above and the equation:

$$\text{mean rate} = \frac{\text{concentration difference}}{\text{time difference}}$$

to calculate the mean rate of ethanol formation for **maltose** between 20 and 50 hours. [2]

mean rate = ... units/hour

(ii) Charlie says that all sugars produce beer with the same ethanol concentration. State whether you agree **and** give **one** reason for your answer. [1]

....

....

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